

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I **Eddu Suchier/192524070** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

4. Suspicious Digital Asset Investigation

a. Analyze how investigators could detect the presence of hidden data within the image.

Step 1: Perform Visual Inspection

Investigators first compare the suspicious company logo with the original image. They look for unusual distortions, changes in image quality, or unexpected differences in file size that may indicate hidden data.

Step 2: Examine Image Metadata

Next, investigators analyze the image metadata (EXIF data) using forensic tools such as **ExifTool**. Metadata may reveal editing software, modification dates, or other suspicious information suggesting the image has been altered.

Step 3: Perform Statistical Analysis

Investigators conduct steganalysis by examining pixel values, Least Significant Bit (LSB) patterns, and image histograms. Unusual statistical patterns may indicate that secret information has been embedded.

Step 4: Use Steganography Detection Tools

Specialized forensic tools such as **StegExpose**, **StegSolve**, or **OpenStego** are used to detect or extract hidden data from the image.

By combining visual inspection, metadata analysis, statistical analysis, and steganography detection tools, investigators can accurately determine whether confidential information has been concealed within the image.

b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Technique 1: Encrypt the Data Before Embedding

Before hiding the message inside the image, encrypt it using **AES-256**. Even if investigators extract the hidden data, they cannot read it without the correct decryption key.

Technique 2: Use Adaptive or Randomized Steganography

Instead of embedding data in fixed pixel locations, use randomized or adaptive embedding techniques. This reduces detectable patterns and makes steganalysis much more difficult.

Using **AES-256 encryption** together with **adaptive steganography** provides stronger confidentiality and significantly reduces the chances of hidden data being detected or extracted.

5. Legacy Encryption in Corporate Communication

a. Determine the most probable original message by reversing the encryption process.

Step 1: Identify the Encryption Method

The ciphertext

WECRLTEERDSOEFEAOCAVDEN

matches the pattern of a **3-Rail Rail Fence Cipher**, a transposition cipher that rearranges the positions of letters without changing them.

Step 2: Reverse the Rail Fence Pattern

The ciphertext is placed back into the three-rail zigzag arrangement used during encryption. Reading the letters in the correct sequence reconstructs the original message.

Step 3: Recover the Original Message

The decrypted plaintext is:

"WE ARE DISCOVERED. FLEE AT ONCE."

The original message obtained after reversing the 3-Rail Rail Fence Cipher is:

"WE ARE DISCOVERED. FLEE AT ONCE."

b. Critically evaluate why such a method is vulnerable in modern communication systems.

Step 1: No Strong Encryption

The Rail Fence Cipher only rearranges the positions of letters instead of encrypting them with a secure mathematical algorithm.

Step 2: Easy to Break

Since there are only a few possible rail arrangements, attackers can quickly recover the original message using brute-force methods or cryptanalysis.

Step 3: Unsuitable for Modern Communication

The cipher provides no protection against modern cyberattacks and lacks important security features such as strong encryption, authentication, and integrity verification.

The Rail Fence Cipher is vulnerable because it uses a simple transposition method that is easy to reverse. It cannot provide the level of security required for protecting confidential corporate communications.

c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Step 1: Replace the Legacy Encryption

Replace the Rail Fence Cipher with a modern encryption algorithm such as **AES-256**.

Step 2: Secure the Communication

AES-256 encrypts the entire message using a strong 256-bit secret key, making it extremely difficult for attackers to decrypt without authorization.

Step 3: Improve Overall Security

The encrypted message can also be transmitted through secure communication protocols such as **TLS** to provide additional protection during transmission.

Replacing the legacy Rail Fence Cipher with **AES-256 encryption** greatly improves confidentiality by protecting messages against brute-force attacks and modern cryptographic threats, making it suitable for secure corporate communication.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

